

Z-residuals for Diagnosing Bayesian Models

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Section 1

Introduction

Residual diagnostics plays very important roles in frequentist statistics:

- **Visualize** the normality of residuals with graphical tools, such as scatterplot and QQ plot.
- **Quantify** the inadequacy of a fitted model using overall goodness-of-fit (GOF) and other diagnostic tests to .
- **Discover** the problems of a fitted model, such as family misspecification, non-linear covariate effect, existence of outliers, and many more.

Our goal of model diagnostics is to figure out how to improve a model, more than just to confirm the good fit of a model.

- Residual diagnostics is not a common practice in Bayesian statistics
- Recently, Z-residuals have been developed to assess the adequacy of various frequentist models, such as survival models and count regression models.
- In this talk, we show how to define Z-residuals for diagnosing Bayesian models with Bayesian hurdle models as an example.

Section 2

Hurdle Models

Hurdle Models

- Hurdle models can be viewed as a two-component mixture model consisting of
 1. a point mass at zero ($\delta_0(\cdot)$), and
 2. a zero-truncated count distribution for the positive observations.
- The probability mass function (PMF) of hurdle models is defined as

$$Pr(Y_i = y_i) = \begin{cases} \pi_i^0, & y_i = 0, \\ (1 - \pi_i^0) \frac{p(y_i; \mu_i)}{1 - p(y_i = 0; \mu_i)}, & y_i > 0, \end{cases} \quad (1)$$

where π_i^0 is the proportion of zeros, $p(y_i; \mu_i)$ represents a PMF for a regular count distribution with a vector of parameters μ_i , and $p(y_i = 0; \mu_i)$ is the distribution evaluated at zero.

Hurdle Models (cont'd)

Examples of Hurdle Models

- Hurdle negative binomial (HNB) models are defined with the following PMF:

$$Pr(Y_i = y_i) = \begin{cases} \pi_i^o, & y_i = 0, \\ \frac{1 - \pi_i^o}{1 - \left(\frac{k}{\mu_i + k}\right)^k} \frac{\Gamma(y_i + k)}{\Gamma(k) y_i!} \left(\frac{\mu_i}{\mu_i + k}\right)^{y_i} \left(\frac{k}{\mu_i + k}\right)^k, & y_i > 0, \end{cases} \quad (2)$$

where μ_i is the mean parameter, k is the dispersion parameter of NB distribution.

- Hurdle Poisson models:

$$Pr(Y_i = y_i) = \begin{cases} \pi_i^o, & y_i = 0, \\ (1 - \pi_i^o) \frac{e^{-\mu_i} \mu_i^{y_i} / y_i!}{1 - e^{-\mu_i}}, & y_i > 0, \end{cases} \quad (3)$$

where μ_i is the mean parameter of Poisson distribution.

- The parameters of zero and count components are linked to covariates:
 - Count model: $\log(\mu_i) = \mathbf{x}_i^T \boldsymbol{\alpha}$
 - Zero model: $\text{logit}(\pi_i^o) = \mathbf{z}_i^T \boldsymbol{\beta}$
 - where μ_i is the mean, π_i^o is the probability of zeros, $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are regression coefficients for the covariates $\mathbf{x}_i$ and $\mathbf{z}_i$.

Section 3

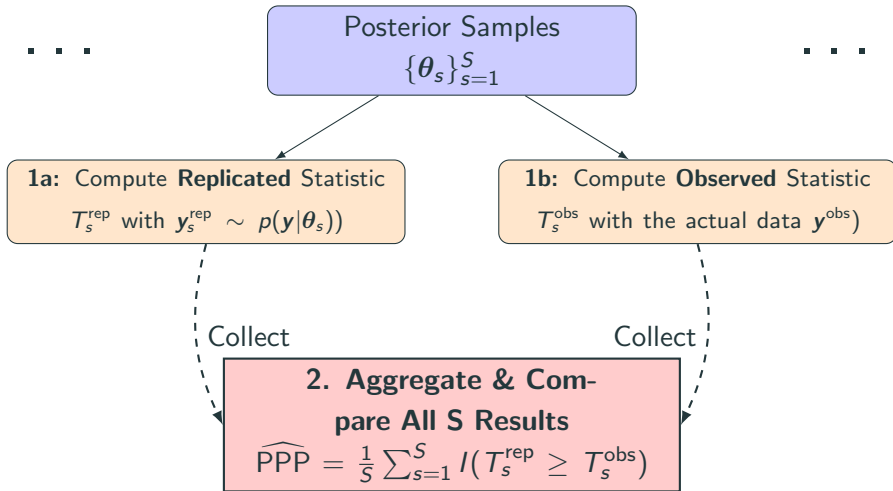
Review of Posterior Predictive Checks

General Description of Posterior Predictive P-value (PPP)

- **What is it?** The Posterior Predictive p-value (PPP) is a goodness-of-fit measure to check if a model is consistent with the observed data, denoted by $\mathbf{y}_{\text{obs}}$.
- **Idea:** We compare the observed data to replicated data generated from the posterior predictive model. If the model fits well, the replicated data should look similar to the observed data.
- Define a **discrepancy statistic**, $T(\mathbf{y}, \boldsymbol{\theta})$, to summarize a specific feature of the data, $\mathbf{y}$. This can be a function of the data alone (e.g., sample mean, $T(\mathbf{y})$) or also depend on the model parameters, $\boldsymbol{\theta}$.
- **Posterior Predictive P-value (PPP):** It is the probability that $T(\mathbf{y}, \boldsymbol{\theta})$ computed on replicated data ($\mathbf{y}_{\text{rep}}$) is more extreme than $T(\mathbf{y}, \boldsymbol{\theta})$ computed on the observed data ($\mathbf{y}_{\text{obs}}$).

$$\text{PPP} = P(T(\mathbf{y}_{\text{rep}}, \boldsymbol{\theta}) > T(\mathbf{y}_{\text{obs}}, \boldsymbol{\theta}) | \mathbf{y})$$

Computing Procedure



Difficulty in Choosing Discrepancy Statistics

- The choice of a test statistic $T(\mathbf{y}, \theta)$ is subjective, depending on what possible problems in the model that one suspects to exist.
- A model might pass checks with simple statistics (e.g., mean, variance) yet fail to capture other critical features of the data. This can create a false sense of “good” model. A “good” p-value for one statistic can mask other significant misfit *that is unknown to the modeler*.

Non-uniformity of Posterior Predictive P-values

- The data is used twice: 1) to fit the model, and 2) to test the model. The posterior predictive p-values are non-uniformly distributed with a bias towards 0.5.
- The bias results in a conservative test with a significant loss of statistical power to detect genuine model misspecifications.

Section 4

Z-residuals

Randomized Predictive P-values (RPP) given True Parameters

The postulated model for $y_i|\mathbf{x}_i, \boldsymbol{\theta}$:

- The PMF is $p_i(y_i|\boldsymbol{\theta})$
- The survival function is $S_i(y_i|\boldsymbol{\theta})$

Predictive p-values for an observed y_i^{obs} :

- The **middle-value predictive p-value (MPP)**:

$$\text{mpp}_i(y_i^{\text{obs}}|\boldsymbol{\theta}) = S_i(y_i^{\text{obs}}|\boldsymbol{\theta}) + 0.5 \times p_i(y_i^{\text{obs}}|\boldsymbol{\theta}). \quad (4)$$

- The **randomized predictive p-value (RPP)** is defined as:

$$\text{rpp}_i(y_i^{\text{obs}}|\boldsymbol{\theta}) = S_i(y_i^{\text{obs}}|\boldsymbol{\theta}) + U_i \times p_i(y_i^{\text{obs}}|\boldsymbol{\theta}) \quad (5)$$

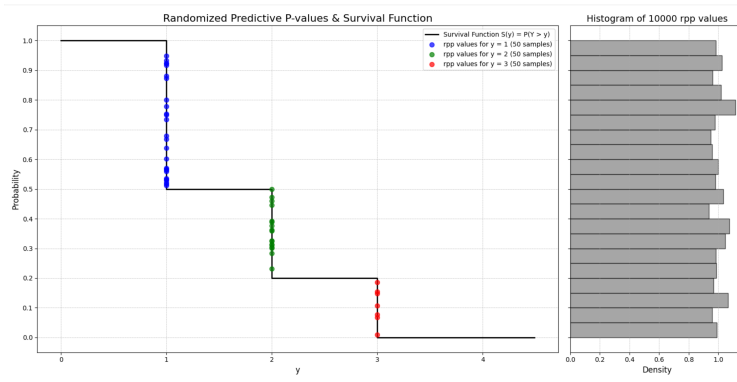
where $U_i \sim \text{Unif}(0, 1)$.

- Under the true model,

$$\text{rpp}_i(y_i^{\text{obs}}|\boldsymbol{\theta})|\mathbf{x}_i \sim \text{Unif}(0, 1) \quad (6)$$

Illustration of the Uniformity of RPPs

Illustration:



Proof: The uniformity of RPPs can be easily proved with elementary probability theory as long as $S(y_i|\theta)$ and $p(y_i|\theta)$ are the true models for y^{obs} .

Transforming RPPs to Normally Distributed Z-residuals

- To compute the **Z-residuals**, we can transform $\text{rpp}_i(y_i^{\text{obs}}|\boldsymbol{\theta})$ with the negative normal quantile:

$$z_i(y_i^{\text{obs}}|\boldsymbol{\theta}) = -\Phi^{-1}(\text{rpp}_i(y_i^{\text{obs}}|\boldsymbol{\theta})), \quad (7)$$

where $\Phi(\cdot)$ is the cumulative distribution function (CDF) of a standard normal distribution.

- Under the true model, the Z-residual is distributed as standard normal.

$$z_i(y_i^{\text{obs}}|\boldsymbol{\theta}) \mid \mathbf{x}_i \sim N(0, 1) \quad (8)$$

Posterior Z-residuals

- The **posterior RPP** $\text{rpp}_i^{\text{post}}(y_i^{\text{obs}})$ is defined as

$$\text{rpp}_i^{\text{post}}(y_i^{\text{obs}}) = \int \text{rpp}_i(y_i^{\text{obs}}|\theta)P(\theta|\mathbf{y}^{\text{obs}})d\theta \quad (9)$$

- The posterior RPP is computed given the Markov Chain Monte Carlo (MCMC) samples, and denoted by $\widehat{\text{rpp}}_i^{\text{post}}(y_i^{\text{obs}})$:

$$\widehat{\text{rpp}}_i^{\text{post}}(y_i^{\text{obs}}) = \frac{\sum_{t=1}^T \text{rpp}_i(y_i^{\text{obs}}|\boldsymbol{\theta}^{(t)})}{T}. \quad (10)$$

- The **posterior RPP Z-residual** can be defined as

$$z_i^{\text{post}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{rpp}}_i^{\text{post}}(y_i^{\text{obs}})). \quad (11)$$

- The posterior MPP and posterior MPP Z-residuals can be defined in the same way.
- We will occasionally use ‘Post.’ to abbreviate the term ‘posterior’.

- The **cross-validators RPP**, $\text{rpp}_i^{\text{CV}}(y_i^{\text{obs}})$, is defined as

$$\text{rpp}_i^{\text{CV}}(y_i^{\text{obs}}) = \int \text{rpp}_i(y_i^{\text{obs}}|\boldsymbol{\theta})P(\boldsymbol{\theta}|\mathbf{y}_{-i}^{\text{obs}})d\boldsymbol{\theta} \quad (12)$$

$$= S_i(y_i^{\text{obs}}|\mathbf{y}_{-i}^{\text{obs}}) + U_i \times p_i(y_i^{\text{obs}}|\mathbf{y}_{-i}^{\text{obs}}) \quad (13)$$

- **Uniformity of CV RPP**

It can be seen easily that

$$\text{rpp}_i^{\text{CV}}(y_i^{\text{obs}})|\mathbf{x}_i \sim \text{Unif}(0, 1) \quad (14)$$

as long as the prior for $\boldsymbol{\theta}$ and the distribution for $y_i|\boldsymbol{\theta}$ are both correctly specified.

The proof is the same as that for the uniformity of $\text{rpp}_i(y_i^{\text{obs}}|\boldsymbol{\theta})$.

Importance-sampling CV Z-residuals

- The ISCV RPP is computed given the MCMC samples $\{(\boldsymbol{\theta}^{(t)}); t = 1, \dots, T\}$, and denoted by:

$$\widehat{\text{rpp}}_i^{\text{ISCV}}(y_i^{\text{obs}}) = \frac{\sum_{t=1}^T [\text{rpp}_i(y_i^{\text{obs}}|\boldsymbol{\theta}^{(t)})w_i^{\text{IS}}(\boldsymbol{\theta}^{(t)})]/T}{\sum_t^T w_i^{\text{IS}}(\boldsymbol{\theta}^{(t)})/T}, \quad (15)$$

where $w_i^{\text{IS}}(\boldsymbol{\theta})$ is the ratio:

$$w_i^{\text{IS}}(\boldsymbol{\theta}) = \frac{P_{\text{post}(-i)}(\boldsymbol{\theta}|y_{-i}^{\text{obs}})}{P_{\text{post}}(\boldsymbol{\theta}|y_{1:n}^{\text{obs}})} \times \frac{C_2}{C_1} = \frac{1}{P_y(y_i^{\text{obs}}|\boldsymbol{\theta})} \quad (16)$$

- The **ISCV RPP Z-residual** can be defined as

$$z_i^{\text{ISCV}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{rpp}}_i^{\text{ISCV}}(y_i^{\text{obs}})). \quad (17)$$

- Same to way define the ISCV MPP Z-residuals.

List of Four Z-residuals

	RPP	MPP
Post.	$z_i^{\text{post-rpp}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{rpp}}_i^{\text{post}}(y_i^{\text{obs}}))$	$z_i^{\text{post-mpp}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{mpp}}_i^{\text{post}}(y_i^{\text{obs}}))$
ISCV	$z_i^{\text{ISCV-rpp}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{rpp}}_i^{\text{ISCV}}(y_i^{\text{obs}}))$	$z_i^{\text{ISCV-mpp}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{mpp}}_i^{\text{ISCV}}(y_i^{\text{obs}}))$

Table 1: Four types of Z-residuals

Section 5

Simulation Study I: Detection of Mis-specified Distribution Family

Subsection 1

Data Generation and Model Fitting

Data Generation from HNB

The data is generated as follows:

- A covariate: $X_i \sim N(0, 1)$.
- Five noises: $Z_{i,j} \sim N(0, 1), j = 1, \dots, 5$.
- Zero model: $\text{logit}(\pi_i^o) = \beta_0 + \beta_1 X_i$.
- Count model: $\log(\mu_i) = \alpha_0 + \alpha_1 X_i$.
- The zero indicator: $O_i \sim \text{Binom}(1, \pi_i^o)$.
- The count response: $y_i = (1 - O_i) \times C_i$,
where $C_i \sim \text{Truncated-NB}(\mu_i, \gamma, \pi)$, $\gamma = 4$ is the dispersion parameter,
and $\pi = \gamma/(\gamma + \mu_i)$ is the probability.
- Regression coefficients: $\alpha_0 = 2, \alpha_1 = 8, \beta_0 = -1, \beta_1 = -1$.
- The above steps are repeated for all simulated datasets. In total, we generate 600 response variables from the HNB model with sample sizes of $n = 50, 100, 200$ and zero proportions of $p = 30\%$ and 60% .

Fitting HNB and NB Models

We fit two candidate models to the simulated datasets utilizing the function `brm()` from R package `brms`. The models are fitted as listed below:

- **HNB Model (true model):**

$$\begin{aligned}y_i &\sim \text{HNB}(\mu_i, \gamma, \pi_i^o), \\ \log(\mu_i) &= \alpha_0 + \alpha_1 X_i + \sum_{j=1}^5 a_j Z_{i,j}, \\ \text{logit}(\pi_i^o) &= \beta_0 + \beta_1 X_i + \sum_{j=1}^5 b_j Z_{i,j};\end{aligned}$$

- **NB Model (wrong model):**

$$\begin{aligned}y_i &\sim \text{NB}(\mu_i, \gamma), \\ \log(\mu_i) &= \alpha_0 + \alpha_1 X_i + \sum_{j=1}^5 a_j Z_{i,j};\end{aligned}$$

Subsection 2

Z-residual Diagnostics

Posterior RPP vs. Posterior MPP Z-residual: Scatter Plot

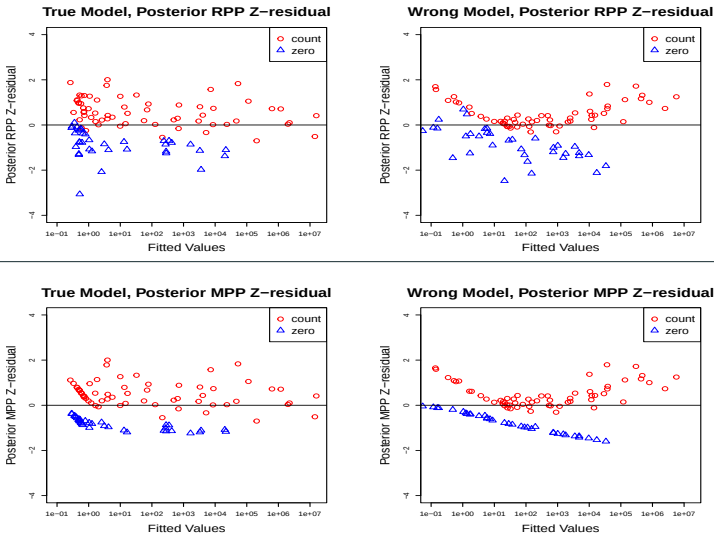


Figure 1: Posterior RPP Z-residuals and posterior MPP Z-residuals vs. covariate for HNB and NB models when the dataset is simulated from HNB ($n = 100, p = 30\%$).

Posterior RPP vs. Posterior MPP Z-residual: Q-Q Plot

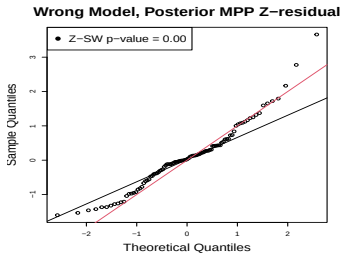
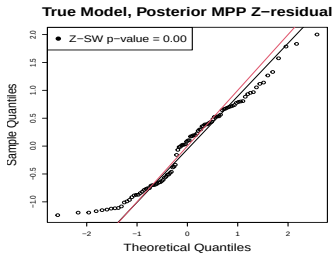
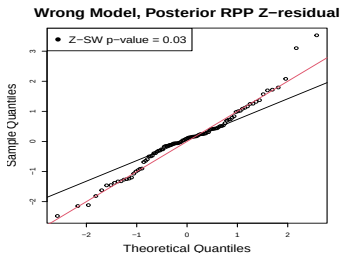
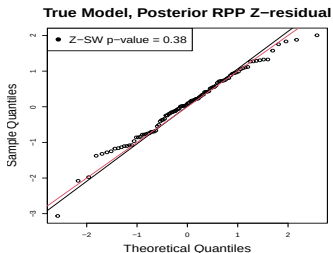


Figure 2: Q-Q plots for posterior RPP Z-residuals vs. posterior MPP Z-residuals for HNB and NB models when the dataset is simulated from HNB ($n = 100, p = 30\%$).

Subsection 3

Replicated Studies for Estimating Type-I Error Rates and Powers

What We Are Comparing

- Posterior Z-residuals vs. ISCV Z-residuals with SW test
- MPP Z-residuals vs. RPP Z-residuals with SW test
- Posterior Predictive P-values with the χ^2 discrepancy measure:

$$T(\mathbf{y}, \boldsymbol{\theta}) = \sum_{i=1}^n \frac{(y_i - E[y_i|\boldsymbol{\theta}])^2}{\text{Var}(y_i|\boldsymbol{\theta})}$$

This discrepancy measure is particularly good at capturing the over-dispersion in the data compared to the fitted model.

The Distributions of the P-values

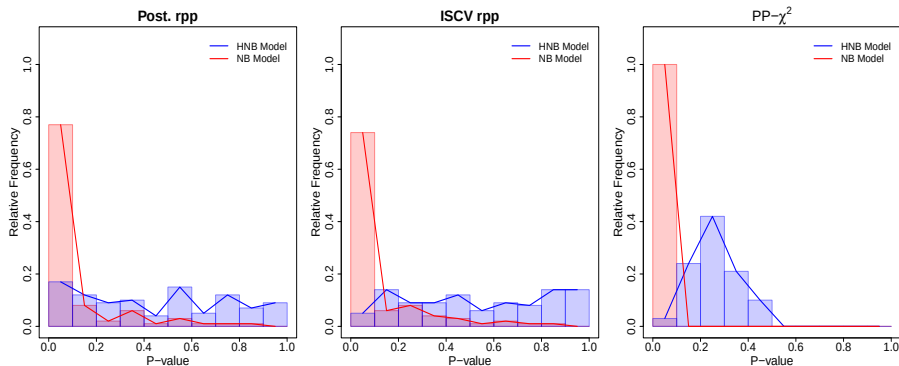


Figure 3: Histograms of P-values based on Shapiro-Wilk (SW) tests of Z-residuals vs. posterior predictive χ^2 when the dataset is simulated from HNB. The number of replications is 100. The zero proportion is $p = 30\%$.

Model Rejection Rates

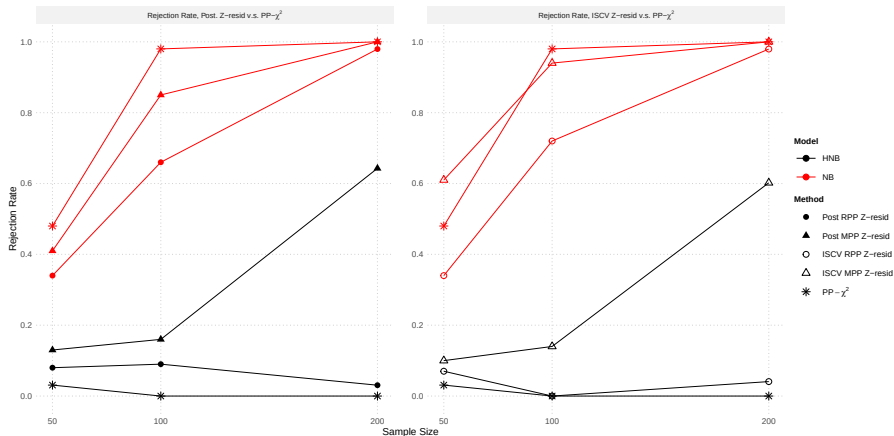


Figure 4: Model rejection rates based on Shapiro-Wilk (SW) tests of RPP vs. MPP Z-residuals when the dataset is simulated from HNB. A model is rejected when the test p -value is smaller than 5%. The number of replications is 100. The zero proportion is $p = 30\%$.

Section 6

Simulation Study II: Detection of Wrong Covariate Functional Forms

Subsection 1

Data Generation and Model Fitting

Data Generation and Model Fitting

We add the **logarithmic covariate effect** to $X_{i,1}$ in the count model.

- Two covariates: $X_{i,1} \sim \text{Unif}(0, 15)$, $X_{i,2} \sim \text{Binom}(1, 0.5)$.
- Five noises: $Z_{i,j} \sim N(0, 1), j = 1, \dots, 5$.
- Regression coefficients:
 $\alpha_0 = -1, \alpha_1 = 6, \alpha_2 = 1, \beta_0 = -1, \beta_1 = -1, \beta_2 = -5$.
- Count model: $\log(\mu_i) = \alpha_0 + \alpha_1 \log(X_{i,1}) + \alpha_2 X_{i,2}$.
- Zero model: $\text{logit}(\pi_i^0) = \beta_0 + \beta_1 X_{i,1} + \beta_2 X_{i,2}$.
- The zero indicator: $O_i \sim \text{Binom}(1, \pi_i^0)$.
- The count response: $y_i = (1 - O_i) \times C_i$,
where $C_i \sim \text{Truncated-NB}(\mu_i, \gamma, \pi)$, $\gamma = 4$ is the dispersion parameter,
and $\pi = \gamma / (\gamma + \mu_i)$ is the probability.
- The steps are repeated for all simulated datasets. We generate 600 response variables from the logarithmic HNB model with sample sizes of $n = 50, 100, 200$ and zero proportions of $p = 30\%$.

- **Logarithmic HNB Model (true model):**

$$y_i \sim \text{HNB}(\mu_i, \gamma, \pi_i^o),$$

$$\log(\mu_i) = \alpha_0 + \alpha_1 \log(X_{i,1}) + \alpha_2 X_{i,2} + \sum_{j=1}^5 a_j Z_{i,j},$$

$$\text{logit}(\pi_i^o) = \beta_0 + \beta_1 X_{i,1} + \beta_2 X_{i,2} + \sum_{j=1}^5 b_j Z_{i,j};$$

- **Linear HNB Model (wrong model):**

$$y_i \sim \text{HNB}(\mu_i, \gamma, \pi_i^o),$$

$$\log(\mu_i) = \alpha_0 + \alpha_1 X_{i,1} + \alpha_2 X_{i,2} + \sum_{j=1}^5 a_j Z_{i,j},$$

$$\text{logit}(\pi_i^o) = \beta_0 + \beta_1 X_{i,1} + \beta_2 X_{i,2} + \sum_{j=1}^5 b_j Z_{i,j}.$$

Subsection 2

Z-residual Diagnostics

Posterior RPP Z-residual: Scatter Plot

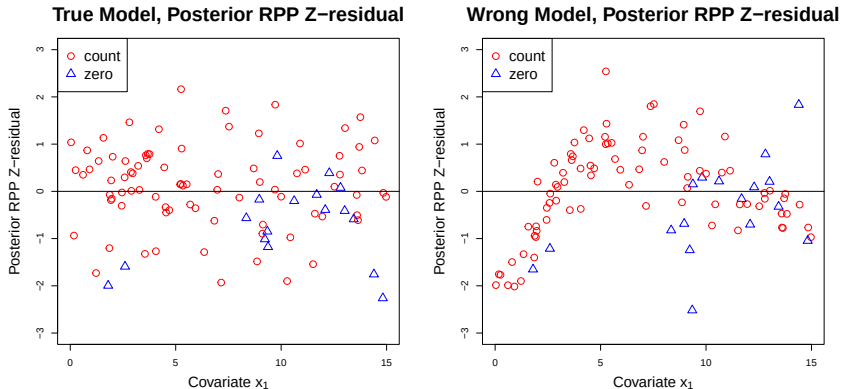


Figure 5: Posterior RPP Z-residuals vs. covariate $X_{i,1}$ for logarithmic HNB and linear HNB models when the dataset is simulated from logarithmic HNB ($n = 100, p = 30\%$).

Posterior RPP Z-residual: Q-Q Plot

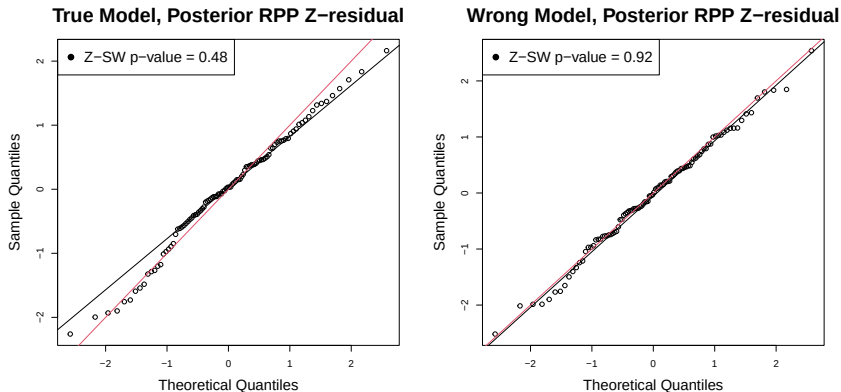
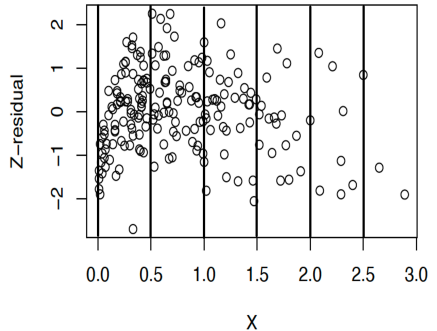
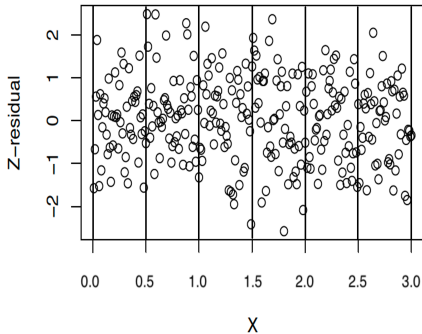


Figure 6: Q-Q plots for posterior RPP Z-residuals for logarithmic HNB and linear HNB models when the dataset is simulated from logarithmic HNB ($n = 100$, $p = 30\%$).

ANOVA with Partitioned Z-residuals for Capturing Mean Trend

Figure 7: An illustrative plot showing how to construct the non-homogeneity test with Z-residuals: dividing Z-residuals by a covariate or linear predictor (LP) with equally-spaced interval, then testing the equality of the means of grouped residuals. We apply the F-test in ANOVA to test the equality of the means of grouped Z-residuals.



Box Plot for ANOVA Tests

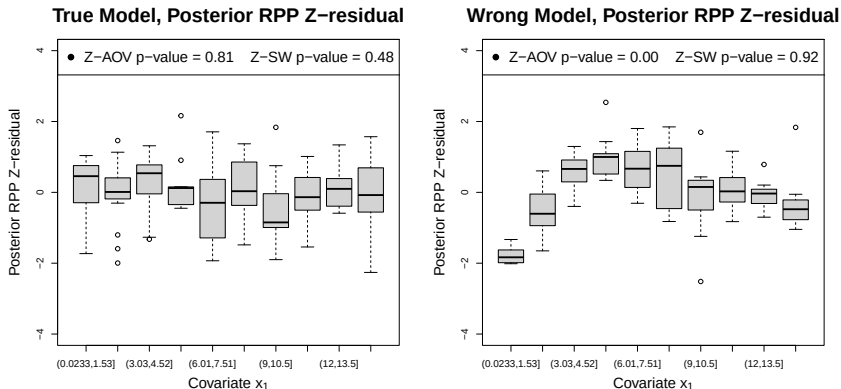


Figure 8: Box plots of posterior RPP Z-residuals for logarithmic HNB and linear HNB models when the dataset is simulated from logarithmic HNB ($n = 100, p = 30\%$). The p -values for the SW test and ANOVA test are shown on the top.

Subsection 3

Replicated Studies for Estimating Type-I Error Rates and Powers

What We Are Comparing

- SW vs. ANOVA Tests Applied to Posterior RPP Z-residuals
- Z-residuals vs. Posterior Predictive Checks with χ^2 Statistic

The Distributions of the P-values

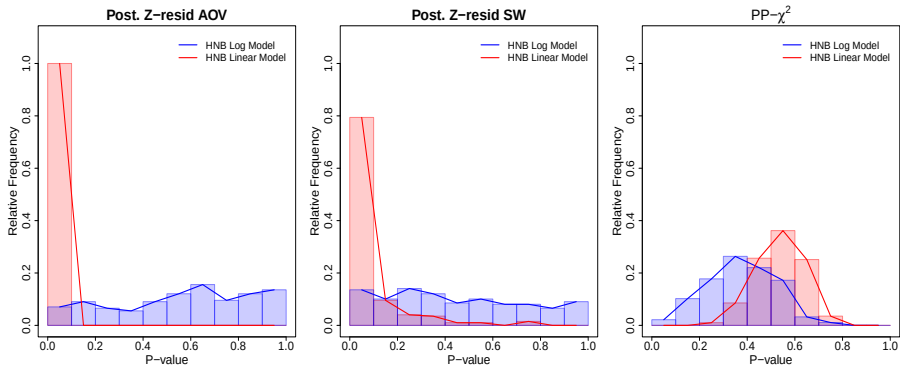


Figure 9: Histograms of P-values based on Shapiro-Wilk (SW) tests of Z-residuals vs. ANOVA tests of Z-residuals vs. posterior predictive χ^2 when the dataset is simulated from HNB. The number of replications is 200. The zero proportion is $p = 30\%$.

Model Rejection Rates

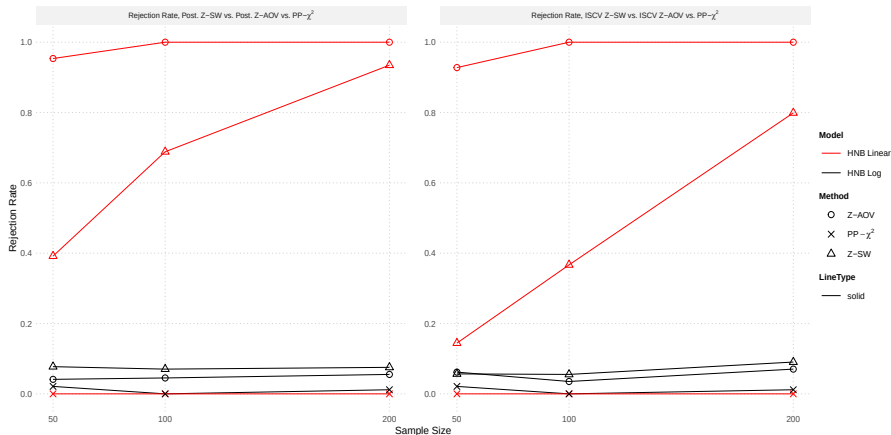


Figure 10: Model rejection rates of RPP Z-residual-based SW (Z-SW) Tests versus RPP Z-residual-based ANOVA (Z-ANOVA) Tests when the dataset is simulated from logarithmic HNB. A model is rejected when the test p -value is smaller than 5%. The number of replications is 600. The zero proportion is $p = 30\%$.

Section 7

Real Data Application

- The data was collected from the National Medical Expenditure Survey (NMES), conducted during 1987 and 1988.
- The dataset analyzed in this study focuses on 4,406 elderly individuals in the United States regarding their healthcare demands.
- The response variable under study is the number of emergency department (ED) visits.
- There are 18 covariates, including demographic factors, socioeconomic variables, and health metrics.

- We fitted hurdle negative binomial (HNB), negative binomial (NB), hurdle Poisson (HP), and Poisson regression models.
- From the previous work, models are reduced to following covariates:
 - Number of chronic conditions (*numchron*)
 - Self-perceived health (*health*, excellent vs. poor; average vs. poor)
 - Limited activities of daily living (*adldiff*, yes vs. no)
 - Number of years of education (*school*)
- The black race (*black*) was significantly associated with increased ED use in the Poisson and HP models but not in the HNB and NB models.

- **HNB Model:**

$$\begin{aligned}y_i &\sim \text{HNB}(\mu_i, \gamma, \pi_i^o), \\ \log(\mu_i) &= \alpha_0 + \alpha_1 * \text{numchron}_i + \alpha_2 * \text{health}_i \\ &\quad + \alpha_3 * \text{adldiff}_i + \alpha_4 * \text{school}_i, \\ \text{logit}(\pi_i^o) &= \beta_0 + \beta_1 * \text{numchron}_i + \beta_2 * \text{health}_i \\ &\quad + \beta_3 * \text{adldiff}_i + \beta_4 * \text{school}_i;\end{aligned}$$

- **NB Model:**

$$\begin{aligned}y_i &\sim \text{NB}(\mu_i, \gamma), \\ \log(\mu_i) &= \alpha_0 + \alpha_1 * \text{numchron}_i + \alpha_2 * \text{health}_i \\ &\quad + \alpha_3 * \text{adldiff}_i + \alpha_4 * \text{school}_i;\end{aligned}$$

- **HP Model:**

$$y_i \sim \text{HP}(\mu_i, \pi_i^o),$$

$$\begin{aligned}\log(\mu_i) = & \alpha_0 + \alpha_1 * \text{numchron}_i + \alpha_2 * \text{health}_i + \alpha_3 * \text{adldiff}_i \\ & + \alpha_4 * \text{school}_i + \alpha_5 * \text{black}_i,\end{aligned}$$

$$\begin{aligned}\text{logit}(\pi_i^o) = & \beta_0 + \beta_1 * \text{numchron}_i + \beta_2 * \text{health}_i + \beta_3 * \text{adldiff}_i \\ & + \beta_4 * \text{school}_i + \beta_5 * \text{black}_i;\end{aligned}$$

- **Poisson Model:**

$$y_i \sim \text{Poisson}(\mu_i),$$

$$\begin{aligned}\log(\mu_i) = & \alpha_0 + \alpha_1 * \text{numchron}_i + \alpha_2 * \text{health}_i + \alpha_3 * \text{adldiff}_i \\ & + \alpha_4 * \text{school}_i + \alpha_5 * \text{black}_i.\end{aligned}$$

Posterior RPP Z-residuals: Q-Q Plots

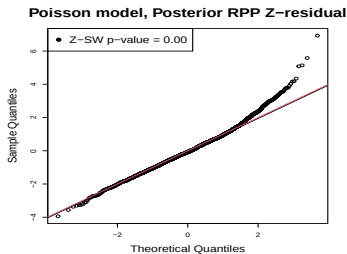
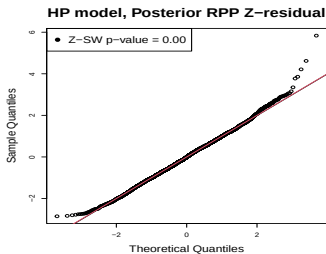
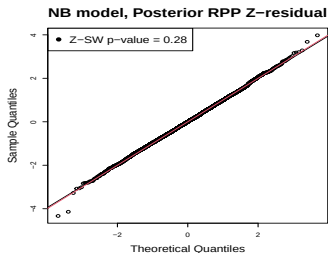
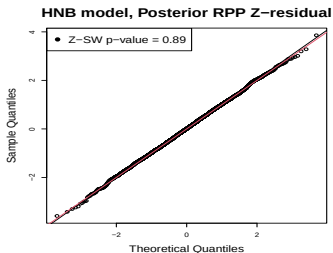
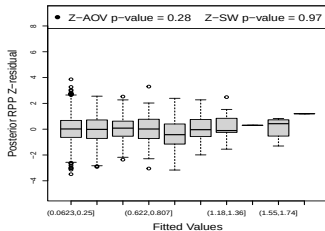
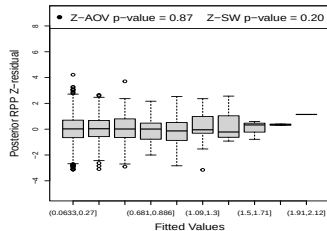


Figure 11: Q-Q plots for Posterior RPP Z-residuals for HNB, NB, HP, and Poisson models.

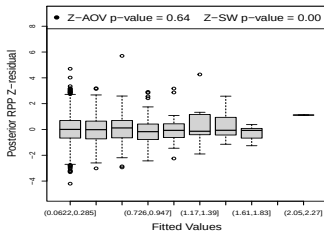
Posterior RPP Z-residuals: Box Plots



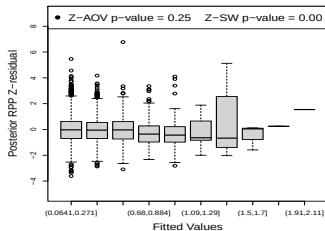
(a) HNB



(b) NB



(c) HP



(d) Poisson

HNB Model Interpretation

Table 2: Estimated regression coefficients, the standard errors (in parentheses), and 95% credible intervals for the HNB models in the real data application

Variables	Count Model		Zero Model	
	Estimate	95% CI	Estimate	95% CI
Intercept	-4.13(2.96)	(-12.16, -1.51)	1.80 (0.13)	(1.55, 2.06)
numchron	0.20 (0.06)	(0.09, 0.31)	-0.21(0.03)	(-0.27, -0.15)
health (poor vs. average)	0.50 (0.50)	(0.10, 0.90)	-0.45(0.11)	(-0.67, -0.23)
health (excellent vs. average)	0.14 (0.49)	(-0.83, 1.09)	0.75 (0.22)	(0.32, 1.18)
adldiff (yes vs. no)	0.40 (0.17)	(0.06, 0.74)	-0.47(0.10)	(-0.65, -0.28)
school	-0.03(0.02)	(-0.07, 0.02)	0.02 (0.01)	(-0.00, 0.04)

- *numchron* (count model): each additional chronic disease is associated with an increase in the expected mean number of ED visits by approximately 22%. This is calculated as $100\% * (e^{0.2} - 1) = 22\%$.
- *numchron* (zero model): with each additional chronic disease and all other variables held constant, the odds of having no ED visits decrease by approximately 19%, calculated as $100\% * (1 - e^{-0.21}) = 19\%$.

Section 8

Conclusions, Future Work, and References

Conclusions and Future Work

- Z-residuals are well-calibrated, powerful, and informative for diagnosing Bayesian models.
- The power of Z-residual diagnosis can be significantly more powerful than posterior predictive p-values.
- The randomization in the RPP is necessary to produce well-calibrated diagnostic p-values based on Z-residuals.
- The ISCV Z-residuals may not be better than the posterior Z-residuals.
- Much more research work is needed to further extend and enhance the application of Z-residuals for other misfits, eg., detection of over-fitting, implementation errors, and models with multiple outcome variables, Z-residuals for latent variables.

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